

JAY KOTHARI

Pune · jaikothari10@gmail.com · (+91) 9428698017 · <https://jaykothari.github.io/>

SUMMARY

Professional is holding Master in engineering with specialization in Embedded system, having work experience of 10 years.

EXPERIENCE

Incedo (<https://www.incedoinc.com>)

Pune, Maharashtra, India

Senior Technical Lead

Oct 2022 - Present

- Incedo is product development partner with <https://www.a10networks.com/products/>. Working in system platform development team. Main responsibility of team is to work on : ACOS <https://www.a10networks.com/solutions/advanced-core-operating-system/>.

Wipro (<https://www.wipro.com/>)

Pune, Maharashtra, India

Senior Software Engineer

March 2021 - Oct 2022

- Telematics-platform bringup and operating system services

Visteon Corporation (<https://www.visteon.com/>)

Pune, Maharashtra, India

Embedded software engineer

July 2019 - Feb 2021

- Smart core is single ECU that combines instrument cluster, a head-up display (HUD) and Visteon's infotainment platform into a single domain controller. Worked on Smart core platform bringup.

Matrix Comsec (<https://www.matrixcomsec.com/>)

Vadodara, Gujarat, India

Senior Software Engineer

Sep 2015 - July 2019

- Network video recorders(NVR) are product to record a video stream from CCTV camera over Ethernet. Design and development NVR.

SLTIET (<https://www.sltiet.edu.in/>)

ad-hoc Lecturer EC department

Rajkot, Gujarat, India

June 2012 - June-2013

EDUCATION

GTU PG School (Gujarat Technological University(GTU), Ahmadabad, Gujarat)

M.E. VLSI & Embedded Systems *GPA: 8.1*

June-2013 - June-2015

Atmiya (<https://atmiyauni.ac.in/> , Gujarat Technological University(GTU))

B.E. Electronics and Communication Engineering *GPA: 8.2*

June-2008 - June-2012

SKILLS

Language:	c, c++
OS:	Good understanding of operating system implementations.
Linux:	Linux System Programming., Linux Kernel Programming, Linux Kernel Debugging skill.
Multi-Threading:	Multi-Threaded application design and development.
RTOS:	Worked with Xenomai Real Time driver development(RTDM).
QNX:	Worked on QNX.
git:	Understanding of software development workflow
Embedded linux:	Porting Linux on custom boards. Experience with Yocto
ARM:	Worked on ARMv7/ARMv8 architecture
Video streaming:	Streaming Video/Audio over RTP using control Protocol RTSP. Raw TCP streaming.
DPDK:	Hands on experience with DPDK development and maintenance

PROJECTS

System platform team at Incedo/A10 networks *c/dpdk/linux/git*

- New hardware platform bringup and implementation of sensor monitoring(using open ipmi), watchdog, LED control, eeprom. Communication over I2C,IPMI(create raw ipmi library).
- Responsible for DPDK maintenance and problem solving.
- Implementation of Link time optimization across the ACOS.
- ARM based platform from Ampere in ACOS. Bringup on platform like vThunder and containers.

- Write a Huge page-based memory allocator for ACOS.
- RHEL 9 support for ACOS.
- Support for high range of CPUs on ACOS. Changes requires in middle-ware.
- Maintenance of various platform images generation.
- Bugs solving and maintenance of middle-ware.

Telematics-platform *c/c++/yocto/embedded linux/git*

Platform: Quectel SOC(AG35) and renesas rh850, Its a telematics control unit(TCU) for which our team is responsible for BSP and operating system services.

- Design and develop Persistence Manager using sqlite3 database. Its a client-server model Persistency Manager application.
- GPIO driver for linux kernel.
- Implementation of low power mode for product. Handling synchronization between Vehical controller(VC) and SOC for various mode of product like service watcher mode, RTC sleep, Superlock mode and Update mode.
- Worked on defect and product stabilation activity. Analyze boot time and optimize the boot time using bootchartd.

Smart Core. *c/andriod linux kernel/git/hypervisor/QNX*

URL : <https://www.visteon.com/newsroom/delivering-digital-integrated-cockpit-solution-smartcore/>

Platform : Qualcomm 6155/6145 processor. Quectel module and renesas rh850 on single platform.

OS:QNX as type-2 hypervisor and Android running as guest OS.

Worked in OS/BSP team for porting and bringing up peripherals.

- Bring-up UART and i2c peripherals(like temperature sensor).
- Bring-up TFT display.
- Bring-up touch panel.
- Early display bring-up in QNX.
- Manage various product variant on the single code base.
- Integrate the Qualcomm SDK versions.

Network Video Recorder *c/c++/linux system programming/network programming/*

URL : <https://www.matrixvideosurveillance.com/network-video-recorder.html>

Platform: HiSilicon 3536 processor(ARM Cortex A17) and TI DM8168(ARM Cortex A8).

- Implementation of recording modules in Network Video Recorder Product.
- Implement RTSP Client and RTSP Server using a live555 library.
- Design and implement a Camera initiation module in product.
- Design and implement 4 synchronous playbacks
- Implement decode and display module for Hi3536 processor.
- Design and implement a hardware test application for product in Qt.
- Porting embedded Linux on HI3536 processor.

Implementatation real time linux driver. *c/linux device driver*

Code : https://github.com/JayKothari/i2c_rtdm

Article : <https://www.opensourceforu.com/2015/10/the-xenomai-project-a-linux-based-rtos/>

Platform : Beagle Bone Black with the Cortex A8 processor with 1GHz Frequency Code.

Description : Design, Implement and Test of Xenomai Real Time drivers(RTDM) for the I2C and UART processor controller on Beagle Bone Black. This is was part of the Thesis in Master of engineering.

PERSONAL PROFILE

Date of Birth: 4th Dec 1989
Gender: Male.
Language(s): English, Hindi and Gujarati.
Permanent Address: 1/7 jairaj plot, Palace Road, Rajkot-360001,Gujarat,India.
Nationality: INDIAN.

DECLARATION

I declare that the above information furnished by me is true to the best of my knowledge and will present an original document whenever asked for.